

Math 460: Exam 1

Student Name: _____

Problem

- Given the **training** data with the binary target variable **Survived** and two input categorical variables **Class** and **Sex**. (Notice: ID is not a variable of interest).

ID	Class	Sex	Survived
1	1	0	1
2	1	0	1
3	1	0	1
4	1	1	1
5	1	0	1
6	1	0	1
7	1	0	1
8	3	1	1
9	2	1	0
10	2	1	0
11	3	1	0
12	2	1	0
13	3	0	0
14	1	0	0
15	2	0	0
16	1	1	0

Using Gini Index as the measure for impurity to:

- Grow the **maximum tree** with three leaves (a stopping rule!) on the data. Draw the (diagram of) tree.
 - Notice: If two splits achieve the same maximum impurity gain, you may randomly select either split as the best split.
- Find the misclassification rate on training data of the maximal tree
- Draw all the possible **subtrees**
- Validate the maximal tree and all the subtrees on the following data to select the **optimal tree**.
 - Note: If the chance of Survived is $1/2$, predict **Survived** (Survived=1)

Class	Sex	Survived
1	0	1
1	1	1
3	1	0
2	0	0